The Modern Credit Rating Game: A Micro-Founded Dynamic Model of the Credit Rating Industry

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Abstract

This document outlines a dynamic structural model of the Credit Rating Agency (CRA) industry, solved numerically to find a Rational Expectations Equilibrium. The framework features three core state variables for each CRA: its individual reputation (R_t), its technology level (K_t), and the aggregate level of investor trust in the market (κ_t). The CRA's choice variable has been expanded from a simple leniency number to a set of discrete Strategic Postures, allowing for more complex, state-dependent strategies. These choices, along with technology investment, are made within a stochastic macroeconomic environment (M_t). The CRA's problem is solved using Value Function Iteration (VFI), and the market equilibrium is explored via a large-scale, agent-based forward simulation. This simulation features a heterogeneous population of individual issuers, each uniquely defined by their credit quality, informational opacity, and size, providing a rich environment for analyzing market dynamics.

1 Literature Review

Credit Rating Agencies (CRAs) are foundational pillars of the modern financial architecture, intended to mitigate information asymmetries between issuers and investors (Cantor, 2007). However, their performance, particularly leading up to and during the 2008 Global Financial Crisis (Sy, 2009; White, 2010), revealed deep-seated conflicts of interest and structural flaws, making the industry a subject of intense academic and regulatory scrutiny. This review synthesizes the key theoretical and empirical literature, tracing the evolution of thought from understanding the core conflicts to analyzing the fragile nature of reputation and the complex effects of regulation, thereby motivating the dynamic modeling approach adopted in this study.

1.1 The Foundational Conflict: Issuer-Pays Model and Strategic Behavior

The modern credit rating industry operates predominantly under an "issuer-pays" business model, a structure identified as the primary source of inherent conflicts of interest (Faure-Grimaud, Peyrache, and Quesada, 2009; Pagano and Volpin, 2012). In this model, the CRA's client is the very entity it is supposed to evaluate objectively. The seminal work of Bolton, Freixas, and Shapiro (2012) provides a cornerstone framework, formalizing this conflict and identifying three fundamental sources of distortion: 1) the CRA's incentive to understate risk to attract and retain business, 2) the ability of issuers to engage in "ratings shopping" by soliciting multiple ratings and selectively disclosing the best one, and 3) the existence of a clientele of "trusting" or less sophisticated investors who take ratings at face value, creating demand for potentially inflated ratings.

Subsequent literature has built upon this foundation, exploring the mechanisms through which these conflicts manifest. Issuers may strategically increase asset complexity, widening the variance in potential assessments and thereby increasing the value of shopping for a lenient rating (Skreta and Veldkamp, 2009). The effectiveness of shopping hinges on opacity in the search process; if investors cannot observe that an issuer has solicited multiple ratings, the issuer can exploit the option to cherry-pick the most favorable outcome (Sangiorgi and Spatt, 2017). This issuer strategy prompts a strategic response from CRAs, leading to "rating catering," particularly for complex assets like CDOs, where CRAs may deviate from their own models to match lenient competitors and retain market share (Griffin, Nickerson, and Tang, 2013). The distinction between solicited (issuer-paid) and unsolicited ratings further complicates the landscape, with the former serving a primary "certification" role and the latter potentially acting as a "monitoring" device, though its disciplinary power remains debated (Fulghieri, Strobl, and Xia,

2014). This inherent conflict, amplified by strategic issuer behavior, sets the stage for the central role of reputation.

1.2 Reputation as the Primary, Yet Fragile, Disciplinary Mechanism

Given the structural conflicts, the primary force expected to ensure rating quality is the CRA's concern for its long-term reputation (Mathis, McAndrews, and Rochet, 2009). Theoretically, the discounted future profits derived from a strong reputation should outweigh the short-term gains from issuing lenient ratings. However, a significant body of work reveals this mechanism to be exceedingly fragile and state-dependent, motivating this study's investigation into the dynamic evolution of reputation and market trust.

First, the effectiveness of reputation appears to be pro-cyclical. Bar-Isaac and Shapiro (2013) formalize this, demonstrating that ratings quality tends to deteriorate during economic booms. In good times, high fee income, low baseline default probabilities, and potentially a larger pool of "trusting" investors (Bolton, Freixas, and Shapiro, 2012) reduce the perceived immediate cost of rating inflation, creating "confidence cycles" (Mathis, McAndrews, and Rochet, 2009). The dynamic nature of credit quality itself, captured in Markov chain models of rating transitions (Camanho, Deb, and Liu, 2012), interacts with this cyclicality, making it harder to disentangle true quality shifts from strategic CRA behavior during expansions. This inherent cyclicality highlights the need for models that can endogenously generate such boom-bust dynamics from rational CRA behavior under conditions of changing underlying risk.

Second, competition, often lauded in other industries, can perversely weaken reputational discipline in the CRA market. By eroding the future rents associated with a good reputation, intense competition strengthens the temptation to engage in a "race to the bottom" (Bolton, Freixas, and Shapiro, 2012). Empirical studies provide strong support: Becker and Milbourn (2011) found that increased competition from Fitch led incumbents S&P and Moody's to issue higher, less accurate ratings. Even more strikingly, Baghai and Becker (2020) show that after suffering reputational damage, S&P strategically *eased* its standards to regain market share, suggesting commercial imperatives can override reputational concerns.

Furthermore, the market's perception and reaction to reputation are dynamic. Reputational shocks, such as major rating failures, not only damage the specific CRA but can also erode overall market trust. Bedendo, Cathcart, and El-Jahel (2018) find that after such shocks, investors react more strongly to subsequent downgrades, indicating heightened skepticism and a loss of confidence in the rating system itself. This dynamic motivates our model's inclusion of an aggregate investor trust state variable (κ_t) and underscores the importance of modeling the feedback loop between CRA actions and market-wide beliefs.

The limitations of viewing reputation monolithically have spurred the development of more nuanced "double reputation" theories. Recognizing that CRAs serve two distinct audiences with conflicting desires, Bouvard and Levy (2018) and Frenkel (2015) propose models where a CRA cultivates a public reputation for accuracy (with investors) alongside a private reputation for accommodation (with issuers). Frenkel (2015) argues this is particularly prevalent in markets with repeat, strategic issuers (e.g., structured finance). Bouvard and Levy (2018) show this can lead to underinvestment in accuracy, as excessive perceived toughness may drive away issuers. A key unanswered question, however, is whether such a double reputation strategy is sustainable in the long run, particularly if investor trust itself is not static but evolves based on observed market outcomes. This study aims to address this dynamic interplay.

1.3 Regulatory Intervention: Shifting Incentives and Unintended Consequences

The manifest failures of reputation and competition to prevent widespread rating inflation before the 2008 crisis prompted significant regulatory intervention globally (Pagano and Volpin, 2012), creating a natural experiment for examining CRA incentives. Regulatory approaches, however, diverged. The U.S. Dodd-Frank Act adopted a "punitive" approach, significantly increasing legal liability for overly optimistic ratings (Type I errors) while imposing negligible penalties for overly pessimistic ones (Type II errors) (Dimitrov, Palia, and Tang, 2015). In contrast, EU regulations took a more "prescriptive" path, mandating certain behaviors like multiple ratings for structured products (Jones et al., 2022; Pagano and Volpin, 2012).

The U.S. approach, in particular, appears to have generated significant unintended consequences. Numerous studies document a post-Dodd-Frank strategic shift toward "defensive pessimism." Dimitrov,

Palia, and Tang (2015) provide influential evidence, finding lower, less informative ratings and more "false warnings" (Type II errors), consistent with a "reputation hypothesis" where CRAs prioritize avoiding litigation over informational accuracy. This conservatism predates Dodd-Frank, with Alp (2013) documenting stricter standards post-Enron/WorldCom, and Baghai, Servaes, and Tamayo (2014) finding evidence of increased conservatism even before Dodd-Frank, suggesting a longer-term trend perhaps amplified by the Act. Huang, Svec, and Wu (2021) show this shift altered market behavior, reducing demand for third ratings near the investment-grade boundary. The EU's prescriptive approach also led to increased conservatism, though potentially driven more by direct regulatory scrutiny (Jones et al., 2022).

However, the impact of regulation is not uniform. Some studies find conflicting evidence (Toscano, 2020), while Opp, Opp, and Harris (2013) develop a model suggesting regulation's effect depends crucially on *how* ratings are used within the regulatory framework (e.g., for bank capital requirements), arguing that regulatory reliance itself can create incentives for inflation independent of the issuer-pays conflict. This highlights that regulation affects both CRA supply of ratings and investor/regulatory demand for them. Reconciling these potentially conflicting effects, Jankowitsch, Ottonello, and Subrahmanyam (2022) find that asset complexity is a key mediating factor: post-Dodd-Frank improvements in informativeness were concentrated *entirely* in simple assets, with no improvement for complex ones. This suggests that when information acquisition is costly, punitive regulation alone may be insufficient. The conflicting evidence regarding the precise impact of asymmetric penalties highlights the need for formal dynamic models that can isolate the effect of the penalty structure itself on CRA strategy.

1.4 The Enduring Challenge: Issuer Opacity, Complexity, and Technology

The mediating role of asset complexity identified by Jankowitsch, Ottonello, and Subrahmanyam (2022) highlights a fundamental challenge: the informational characteristics of the issuer and its assets. Opaque or complex assets, like structured finance products or firms in inherently less transparent industries (Morgan, 2002), are harder to value accurately (Bolton, Freixas, and Shapiro, 2012). This opacity creates greater scope for strategic behavior by both issuers (ratings shopping (Skreta and Veldkamp, 2009)) and CRAs (catering (Griffin, Nickerson, and Tang, 2013)).

A frontier of the literature explores the issuer's active role in shaping this information environment through "falsification" or strategic manipulation (Perez-Richet and Skreta, 2022). Cohn, Rajan, and Strobl (2024) show that stricter CRA screening might paradoxically *increase* an issuer's incentive to manipulate information. This creates a potential "arms race" dynamic. However, the literature predominantly treats the CRA's ability to combat complexity—its analytical technology and investment in rating quality—as an exogenous parameter or a fixed cost (Krämer and Wiewiorra, 2012). This overlooks a critical dimension of competition and strategic investment, representing a significant gap in understanding. How do CRAs strategically invest in technology (K_t) to navigate the challenges of opacity, and how does this investment interact with their reputational concerns and choice of rating posture? Endogenizing this investment decision connects the CRA literature to dynamic industrial organization models where quality and market structure co-evolve (Fischer, 2015; Hirth, 2014).

1.5 Research Gaps and Contribution of the Present Study

The existing literature, while rich, leaves important gaps, primarily stemming from its reliance on static or limited-period analyses. Key dynamic elements—the long-run evolution of reputation, the co-evolution of CRA strategy and market trust, and strategic investment in technology—remain under-modeled. Conflicting empirical findings on competition and regulation underscore the need for a unified framework capable of capturing dynamic trade-offs and heterogeneous responses. The present study contributes by developing a dynamic, micro-founded structural model of the CRA industry solved in a rational expectations equilibrium to formally investigate the following core research questions and hypotheses:

Endogenous Pro-Cyclicality and Reputational Dynamics. How do boom-bust cycles in rating quality emerge from rational CRA behavior? Building on Bar-Isaac and Shapiro (2013) and Mathis, McAndrews, and Rochet (2009), we move beyond exogenous cyclical shocks.

Hypothesis 1 (Endogenous Pro-cyclicality)**.** *The model's equilibrium generates endogenous pro-cyclicality. A rational, forward-looking CRA will optimally choose more lenient rating postures during prolonged economic booms (when latent risks are masked), leading to a subsequent sharp collapse in reputation* (R_t) *when an economic bust reveals accumulated rating errors.*

The Unintended Consequences of Asymmetric Regulation. Can the shift to asymmetric penalties (like Dodd-Frank) alone explain the observed "defensive pessimism," reconciling conflicting findings (Dimitrov, Palia, and Tang, 2015; Jones et al., 2022; Toscano, 2020; Opp, Opp, and Harris, 2013)?

Hypothesis 2 (Defensive Pessimism)**.** *Simulating a shift from a symmetric to an asymmetric penalty regime (i.e., high penalties for Type I errors, no penalty for Type II errors) will cause the CRA's optimal equilibrium strategy to pivot from lenient/honest postures toward "defensive pessimism." This shift will result in a decrease in Type I errors but a significant increase in Type II errors (false warnings).*

Technology Investment as a Strategic Variable. How do CRAs strategically invest in analytical technology, especially given issuer opacity (Jankowitsch, Ottonello, and Subrahmanyam, 2022) and potential falsification (Cohn, Rajan, and Strobl, 2024), moving beyond models with exogenous quality (Krämer and Wiewiorra, 2012)?

Hypothesis 3 (Technology as a Strategic Complement)**.** *Technology investment emerges as a strategic complement to reputation, not a substitute for it. The model's VFI solution will show that the optimal policy $i^*(S_j)$ is to invest in technology $(i = 1)$ primarily in states of high reputation (R_t). This investment is used to compete for high-margin opaque issuers, while CRAs in low-reputation states will forgo investment and compete on leniency.*

The Sustainability of Double Reputation in an Adaptive Market. Is a "double reputation" strategy (Frenkel, 2015; Bouvard and Levy, 2018) sustainable when investor trust (κ_t) evolves dynamically based on market outcomes (Bedendo, Cathcart, and El-Jahel, 2018)?

Hypothesis 4 (Instability of Double Reputation)**.** *The "double reputation" strategy (i.e., 'Strategic Leniency') is not a stable long-run equilibrium but a transient, pro-cyclical phenomenon. The simulation will show that this strategy is only optimal in high-trust states (κ_t). Its use leads to latent risk accumulation, which, in a bust, triggers a collapse in both individual reputation (R_t) and aggregate trust (κ_t), rendering the lenient strategy unprofitable.*

By endogenizing strategic posture choice, technology investment, individual reputation, and aggregate market trust within a dynamic framework featuring heterogeneous agents, this study provides a computational laboratory to explore these fundamental questions about the modern credit rating game.

2 The Model

The model is set in discrete time, $t = 1, 2, ...$, and consists of two main types of agents: a set $\mathcal{J}$ of J CRAs and a population of N individual corporate issuers.

2.1 Macroeconomic Environment

The **macroeconomic environment** evolves according to a stochastic process, where the state M_t can be one of two possibilities, $M_t \in \{\text{Boom, Bust}\}$. The state evolves according to a first-order Markov transition matrix, $\mathbf{P}$:

$$\mathbf{P} = \begin{pmatrix} p_{B|B} & p_{Bu|B} \\ p_{B|Bu} & p_{Bu|Bu} \end{pmatrix} \tag{1}$$

where $p_{j|i}$ is the probability of transitioning to state j given the current state is i. The primary impact of the macroeconomic state is on the true default probability of issuers. During a Boom, all underlying default probabilities are systematically lower, whereas in a Bust, they are higher.

2.2 Issuer's Problem with Heterogeneous Preferences

The state of each CRA $j \in \mathcal{J}$ at time t is defined by the vector $S_{j,t} = (R_{j,t}, K_{j,t}, \kappa_t, M_t)$, where $R_{j,t}$ is its individual reputation, $K_{j,t}$ is its technology level, and κ_t represents the aggregate level of investor trust. CRAs are the choice-set for the issuers.

The population of **issuers** is heterogeneous. Each issuer $k \in \{1, ..., N\}$ is defined by a unique, time-invariant type vector θ_k:

$$\theta_k = (s_k^*, O_k, Z_k) \tag{2}$$

where $s_k^* \in [0, 1.1)$ is a latent, continuous score representing true credit quality, $O_k \in \{Transparent, Opaque\}$ is the issuer's informational opacity, and $Z_k \in \{Small, Large\}$ is the issuer's size.

2.2.1 The Issuer's Choice: A Utility-Based Approach

The model abstracts away from explicitly modelling end-investors and the pricing of securities. Instead, the benefits an issuer receives are captured in a reduced-form utility function representing advantages like a lower cost of capital, broader market access, and managerial prestige. To understand the forces driving an issuer's choice of CRA, we first deconstruct the issuer's sensitivity to the rating level itself, before building up the full utility function that incorporates the CRA's credibility.

Deconstructing Leniency: Heterogeneous Preferences. A core assumption is that the benefit of a lenient, upgraded rating is not uniform across all firms. The model captures this through a leniency sub-utility function, U_{len}, where an issuer's sensitivity, α_k, is a function of its underlying type $\theta_k = (s_k^*, O_k, Z_k)$. This sensitivity is influenced by both true credit quality (s_k^*) and opacity (O_k):

$$\alpha_k(\theta_k) = \alpha_{\text{base}}(s_k^*) \cdot \Omega(O_k) \tag{3}$$

The base sensitivity, $\alpha_{\text{base}}(s_k^*)$, is highest for issuers on the cusp of investment grade ('BBB', 'BB'), significantly lower for junk-rated issuers ('B', 'CCC'), and zero for high-grade issuers who are indifferent to leniency. This sensitivity is then amplified for 'Opaque' issuers by the multiplier $\Omega(O_k)$, reflecting their heightened desire for a favorable rating to overcome informational frictions:

$$\Omega(O_k) = 1 + (\omega_{amp} - 1) \cdot \mathbb{I}(O_k = \text{Opaque}) \tag{4}$$

Finally, the utility from a given level of leniency, l, exhibits diminishing marginal returns, where the first notch of an upgrade is highly valuable, but subsequent notches are less so. This is modelled using a standard power function:

$$U_{\text{len}}(l, \alpha_k) = \begin{cases} \alpha_k \left(\frac{(1+l)^\phi - 1}{\phi} \right) & \text{if } l \geq 0 \\ \frac{\alpha_k}{3} l & \text{if } l < 0 \end{cases} \tag{5}$$

where $\phi \in (0, 1)$ is the leniency curvature parameter. The combined effects of these mechanisms are illustrated in Figure 1, which shows that opaque, cusp-rated issuers have the most to gain from a lenient rating.

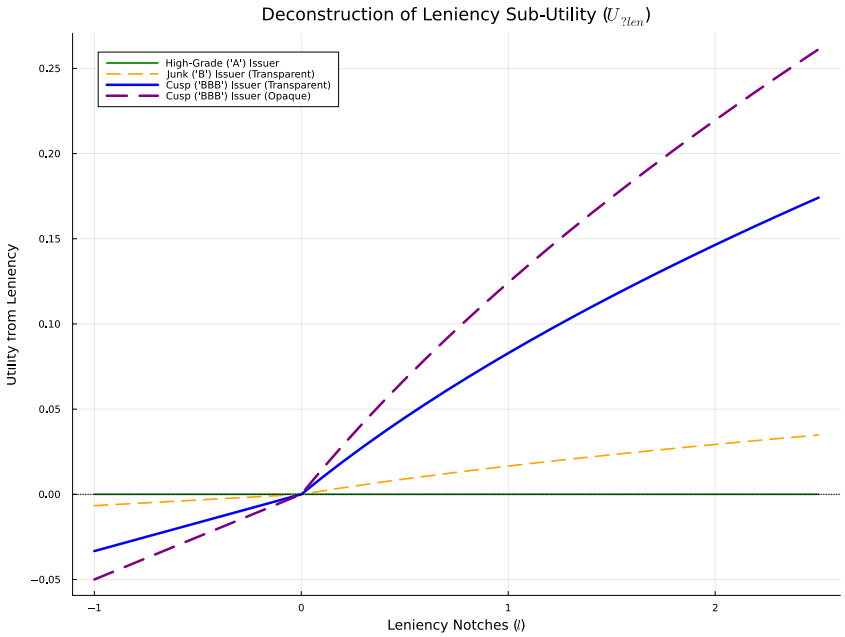


Figure 1: Deconstruction of the Leniency Sub-Utility (U_{len}).

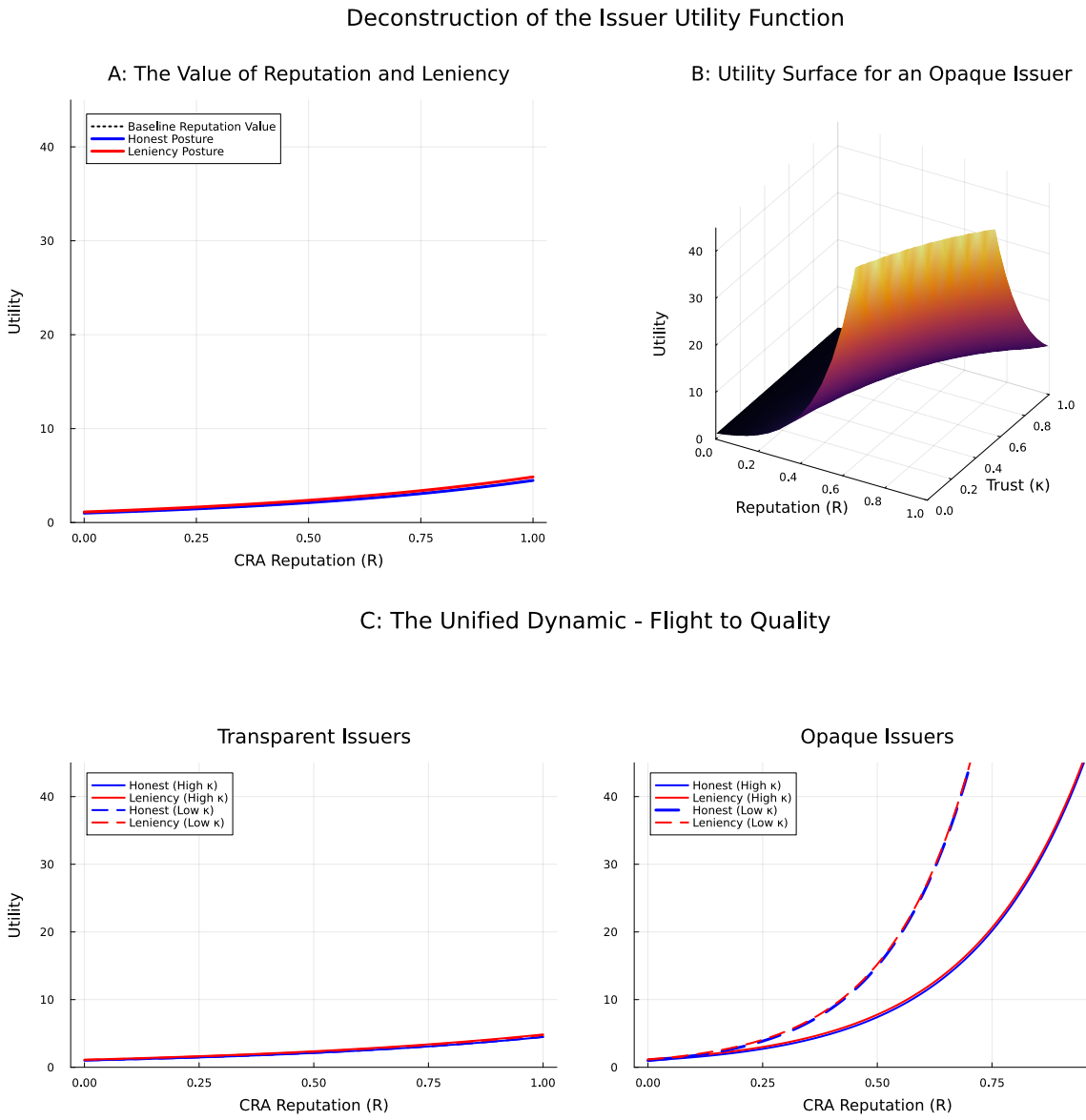


Figure 2: Deconstruction of the Full Issuer Utility Function.

The Role of Reputation and Market Trust. Second, the value of a rating is intrinsically tied to credibility. The model deconstructs this concept into several layers, which are visualized in Figure 2.

The most basic component is the direct "stamp of approval" an issuer receives from being rated by a credible CRA. This baseline utility, shown by the dotted line in Panel A, is an exponential function of the CRA's individual reputation ($R_{j,t}$):

$$U_{\text{base}}(R_{j,t}, \theta_k) = \exp(\gamma_k(\theta_k) R_{j,t}) \tag{6}$$

Adding the Value of Leniency. Issuers also value receiving a more favourable rating. This is captured by the leniency sub-utility, U_{len}, which is highest for "cusp" issuers ('BBB', 'BB') and exhibits diminishing returns. As shown in Panel B, this creates a premium for choosing a CRA with a 'Leniency' posture over an 'Honest' one, at any given level of reputation.

The Influence of Market Trust (κ_t) and Reputation ($R_{j,t}$). The value of a lenient rating is not absolute; it depends on whether the market will accept it. This credibility is captured along two dimensions: aggregate market trust, κ_t, and the CRA's specific reputation, $R_{j,t}$. A lenient rating is most valuable when the market is generally trusting (high κ_t) and when it comes from a credible source (high

$R_{j,t}$). As illustrated in Panel C, a collapse in market trust significantly erodes the utility premium from leniency, as the rating is more likely to be scrutinized.

The Unified Dynamic: Flight to Quality. These forces combine to create a rich, state-dependent utility function. The final component captures a "flight to quality" dynamic that is most pronounced for high-quality, opaque issuers (the "diamonds in the rough"). During a crisis of confidence (low κ_t), the value of a credible signal for these hard-to-value firms soars. The full utility function captures this by adding a three-way interaction between market trust, issuer opacity, and issuer quality to the direct value of reputation:

$$\begin{aligned} U(S_{j,t}, P_{j,t}, \theta_k) = \quad & \underbrace{U_{\text{len}}(...) \cdot (1 + \kappa_{\text{amp}}\kappa_t) \cdot (1 + \gamma_{\text{int}} R_{j,t})}_{\text{Value of Leniency, amplified by Trust and Credibility}} \\ & + \underbrace{\exp\left(\gamma_k(\theta_k) R_{j,t} \cdot (1 + \gamma_{\text{flight}}(1 - \kappa_t) \cdot \mathbb{I}(O_k = \text{Opaque}) \cdot s_k^*)\right)}_{\text{Direct Value of Reputation with Flight-to-Quality Interaction}} \end{aligned} \tag{7}$$

The result of this final interaction is demonstrated in Panel C. For an opaque issuer in a low-trust market (bottom-right), the utility of being certified by a high-reputation 'Honest' CRA soars, surpassing the value of a lenient rating and illustrating the powerful demand for a credible signal during a crisis.

The Logit Choice Framework. Given the utility derived from each CRA, issuers choose whom to hire. To avoid unrealistic "knife-edge" outcomes, the choice is modeled using a standard logit formulation. The probability that issuer k chooses CRA j is its utility relative to all other available options, including an outside option U_{out}:

$$P(\text{choose } j|\theta_k) = \frac{\exp\left(U(R_{j,t}, P_{j,t}, \theta_k)\right)}{\sum_{i \in \mathcal{J}} \exp\left(U(R_{i,t}, P_{i,t}, \theta_k)\right) + \exp(U_{\text{out}})} \tag{8}$$

This formulation ensures that market share is a continuous function of a CRA's reputation and strategic posture.

2.3 The Credit Rating Agency's Dynamic Problem

Each Credit Rating Agency (CRA) $j \in \mathcal{J}$ is a forward-looking agent seeking to maximize the present discounted value of its future after-tax profit stream. Its primary decisions in each period are the choice of a strategic posture $P_{j,t}$ from a discrete set $\mathcal{P}$, and a binary technology investment decision $i_{j,t} \in \{0, 1\}$. This optimization problem is captured by the Bellman equation for the CRA's value function $V(S_{j,t})$:

$$V(S_{j,t}) = \max_{P_{j,t} \in \mathcal{P}, i_{j,t} \in \{0,1\}} \{\Pi_{j,t} - \Psi(R_{j,t}) + \beta \mathbb{E}_t[V(S_{j,t+1})]\} \tag{9}$$

where $\Pi_{j,t}$ is the after-tax profit in period t. The current-period return includes not only profit, but also a non-pecuniary penalty for low reputation, $\Psi(R_{j,t}) = \psi_{rep}(1 - R_{j,t})^2$, which captures intangible costs such as damage to brand value.

The set of available postures $\mathcal{P}$ is {*Conservative, Honest, Uniform Leniency, Aggressive Leniency, Strategic Leniency*}, each corresponding to a specific rule for adjusting an issuer's rating.

The rating adjustment function $l(P, \theta_k)$ is then defined as:

$$l(P, \theta_k) = \begin{cases} -1 & \text{if } P = \text{Conservative} \\ 0 & \text{if } P = \text{Honest} \\ +1 & \text{if } P = \text{Uniform Leniency} \\ +1 \cdot \mathbb{I}_{\text{strategic}}(\theta_k) & \text{if } P = \text{Strategic Leniency} \\ +2 & \text{if } P = \text{Aggressive Leniency} \end{cases} \tag{10}$$

$\mathbb{I}_{\text{strategic}}(\theta_k)$ is the indicator function defined as:

$$\mathbb{I}_{\text{strategic}}(\theta_k) = \begin{cases} 1 & \text{if } O_k = \text{Opaque and } Z_k = \text{Large} \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

The solution to this dynamic programming problem is a set of time-invariant **policy functions**, $(P^*(S_j), i^*(S_j))$. These functions serve as a complete decision-making rule, mapping any possible state S_j that the CRA might find itself in to its optimal strategic actions. The VFI procedure solves for these functions, which are then used by the CRA agents in the forward simulation.

2.3.1 Rating Assessment and Assignment

The core function of a CRA is to assess an issuer's creditworthiness, a process subject to imperfect information. The model conceptualizes an issuer's "true" financial health as a **latent credit quality**, s_k^*. This is a continuous variable representing the firm's fundamental, underlying quality, which is unobservable to any market participant. The familiar discrete rating categories ('AAA', 'AA', etc.) are simply coarse buckets into which this continuous quality is mapped.

Because a CRA cannot observe s_k^* perfectly, it must form an estimate by gathering and analyzing information. This estimate is represented as a **noisy signal**, $\tilde{s}_{k,j,t}$, which is the true latent quality plus a random error term:

$$\tilde{s}_{k,j,t} = s_k^* + \epsilon_{k,j,t} \tag{12}$$

where the error term $\epsilon_{k,j,t}$ is drawn from a normal distribution $\epsilon_{k,j,t} \sim \mathcal{N}(0, \sigma^2_{k,j,t})$.

The Determinants of Assessment Difficulty A key feature of the model is that the difficulty of assessing an issuer is not uniform. Some issuers are inherently harder to analyze than others, and some CRAs are better at the task. This is captured by making the variance of the signal noise heteroscedastic, depending on both the issuer's type and the CRA's characteristics:

$$\sigma^2_{k,j,t} = \sigma^2_{base} \cdot \underbrace{(1 + \omega_{noise} \cdot \mathbb{I}(O_k = \text{Opaque}))}_{\text{Opacity}} \cdot \underbrace{(1 - K_{j,t})}_{\text{Technology}} \cdot \underbrace{(1 + (1 - s_k^*))}_{\text{Quality}} \tag{13}$$

The intuition for each component is as follows:

- **Opacity (O_k):** Issuers with complex structures or less transparent financial reporting are inherently harder to evaluate. The model captures this by increasing the noise variance for issuers marked as 'Opaque'.
- **Technology ($K_{j,t}$):** A CRA can improve its analytical capabilities by investing in technology and expertise. A higher technology level, $K_{j,t}$, allows the CRA to cut through the noise more effectively, directly reducing the variance of its signal.
- **Quality (s_k^*):** It is often more difficult to precisely distinguish between different grades of low-quality or distressed firms than it is to assess high-quality, stable ones. This equation captures that by increasing the noise variance as an issuer's true quality s_k^* decreases, making the speculative-grade space inherently more uncertain.

The Two-Stage Rating Process With this noisy signal in hand, the CRA's rating process proceeds in two distinct stages:

1. **Objective Assessment:** The CRA first translates its numerical estimate $\tilde{s}_{k,j,t}$ into a *perceived* rating category, $\tilde{c}_{k,j,t}$. This can be thought of as the raw, objective output of its internal models.
2. **Strategic Adjustment:** The CRA then applies its chosen strategic posture $P_{j,t}$ to this perceived rating to arrive at the final *assigned* rating, $c'_{k,j,t}$. This step represents the firm's deliberate, strategic overlay on top of its analytical assessment.

2.3.2 Default Realization and Latent Risk

The model's ability to generate pro-cyclical rating quality stems from the mechanism governing how defaults are realized. This mechanism is designed to capture the well-documented phenomenon that the true risk of leniently-rated assets can be hidden during benign economic conditions. The default probability for an issuer k, $p^{\text{default}}_{k,t+1}$, therefore depends on both the macroeconomic state, M_{t+1}, and the rating it was assigned, $c'_{k,j,t}$.

During a `Boom`, an issuer's default probability is based on the benchmark rate, $\pi^*_{c'}$, associated with its potentially inflated *assigned* rating. This is a reduced-form representation of real-world economic

channels where the rating acts as a short-term, self-fulfilling prophecy by altering the firm's economic environment. A higher assigned rating provides tangible survival benefits by: (1) improving access to capital, as many institutional investors are restricted to holding investment-grade debt, thereby lowering the cost of capital for firms at or above this threshold (Kisgen, 2006); (2) enhancing operational stability, as a higher rating signals reliability to customers and suppliers, leading to more favourable trade credit terms (Petersen and Rajan, 1997); and (3) relaxing financial constraints by helping firms avoid violating performance-based debt covenants (e.g., those tied to cash flow or leverage), which can trigger technical defaults or penalties (Sufi, 2009). These channels provide a liquidity buffer that allows a fundamentally weaker firm to meet its obligations, thus masking its underlying risk and making the lenient rating appear accurate.

However, a `Bust` acts as a stress test that can negate these benefits. With probability $\rho_{\text{fragility}}$, this hidden risk is revealed. In this state, the issuer's default probability reverts to one based on its *true* underlying quality, s_k^*. This true default risk is also amplified by the degree of rating inflation it received, capturing the idea that firms who sought and received lenient ratings were likely engaged in riskier activities, making them particularly vulnerable in a downturn. This is formalized as:

$$p_{k,t+1}^{\text{default}} = \begin{cases} \pi_{c'}^* \cdot m_{t+1} & \text{if } M_{t+1} = \text{Boom, or risk is not revealed in a Bust} \\ \pi(s_k^*) \cdot \mu_{\text{inf}}(l_k) \cdot m_{t+1} & \text{if } M_{t+1} = \text{Bust and risk is revealed} \end{cases} \tag{14}$$

where $\pi(s_k^*)$ is the default probability corresponding to the issuer's true quality, and $\mu_{\text{inf}}(l_k)$ is an inflation multiplier that increases with the degree of leniency, l_k, the issuer received. This environmental feature is fully understood by the CRAs, and their decision to grant leniency is therefore a rational, forward-looking gamble on the persistence of the economic boom.

2.3.3 Profits and State Evolution

The CRA's choices and the resulting market outcomes determine its financial performance and the evolution of its state variables. After-tax profit, $\Pi_{j,t}$, is determined by the CRA's expected fee income, $\mathbb{E}[F_{j,t}]$, less a fixed operational cost, ψ, and any technology investment cost, C_{inv}, adjusted for the corporate tax rate, τ:

$$\Pi_{j,t} = (\mathbb{E}[F_{j,t}] - \psi - i_{j,t} \cdot C_{inv})(1 - \tau) \tag{15}$$

The technology level evolves based on the investment decision and a constant rate of decay, d_K, with new investment providing an incremental improvement of Δ_K:

$$K_{j,t+1} = (1 - d_K)K_{j,t} + i_{j,t} \cdot \Delta_K \tag{16}$$

The evolution of reputation, $R_{j,t+1}$, and market trust, κ_{t+1}, are governed by separate learning mechanisms detailed below. This closes the dynamic loop, as decisions and outcomes in period t determine the CRA's state entering period $t + 1$.

2.4 Reputation Dynamics: A Two-Stage Bayesian Update

The reputation update mechanism is designed to capture the procyclicality of detection risk, where rating inflation is easier to conceal in booms and more likely to be exposed in busts. It operates as a two-stage process, incorporating a "zone of inertia" where minor deviations in default performance are dismissed by the market as statistical noise.

2.4.1 Stage 1: Statistical Surprise Check

A reputation update is only triggered if the CRA's performance in any single rating category is sufficiently poor to arouse market suspicion. For each rating category c in its portfolio, the model calculates the expected number of defaults, $\mathbb{E}[k_c]$, and the standard deviation of defaults, σ_k, conditional on the number of firms rated n_c, the benchmark default probability π_c^*, and the macro-state multiplier m_{t+1}. Suspicion is triggered if, for any category c, the actual number of defaults k_c exceeds the expected number by a statistically significant margin, defined by a z-score threshold z:

$$\text{Suspicion triggered if } \exists\ c \text{ s.t. } k_c > \mathbb{E}[k_c] + z \cdot \sigma_k \tag{17}$$

This standard-deviation-based threshold is crucial; it is less sensitive in absolute terms during busts (when $\mathbb{E}[k_c]$ and σ_k are high) and more sensitive during booms, capturing the idea that even a few unexpected defaults in a high-grade category during good times is highly suspicious.

If no category triggers suspicion, the CRA is presumed to be performing as expected. In this case, its reputation evolves according to a slow, upward drift towards 1.0, governed by the reputation drift parameter, ρ_{drift}:

$$R_{j,t+1} = (1 - \rho_{\text{drift}})R_{j,t} + \rho_{\text{drift}} \cdot 1.0 \tag{18}$$

2.4.2 Stage 2: Conditional Bayesian Update

If suspicion is triggered, the market updates its belief about the CRA's type (honest or inflating) using Bayes' rule. The market's prior belief that the CRA is honest is its current reputation, $R_{j,t}$. After observing the portfolio's defaults, $\mathcal{D}_t = \{k_c, n_c\}_{\forall c}$, the market forms a posterior belief by comparing two hypotheses:

- **H_0: The CRA is honest.** The expected default probability for a bond in category c is $p(\text{default}|H_0) = \pi^*_c \cdot m_{t+1}$.
- **H_1: The CRA is inflating by one notch.** The market assumes the bonds rated c are actually of quality $c+1$, so $p(\text{default}|H_1) = \pi^*_{c+1} \cdot m_{t+1}$.

The log-likelihood of observing the defaults under each hypothesis is calculated across the entire portfolio using the binomial probability mass function:

$$\log \mathcal{L}(\mathcal{D}_t|H) = \sum_c \log \text{Binom}(k_c; n_c, p(\text{default}|H)) \tag{19}$$

The posterior probability of the CRA being honest (the updated reputation), $R'_{j,t+1}$, is then given by Bayes' rule:

$$R'_{j,t+1} = P(H_0|\mathcal{D}_t) = \frac{\mathcal{L}(\mathcal{D}_t|H_0)R_{j,t}}{\mathcal{L}(\mathcal{D}_t|H_0)R_{j,t} + \mathcal{L}(\mathcal{D}_t|H_1)(1 - R_{j,t})} \tag{20}$$

Finally, the updated posterior is integrated with the prior reputation using a belief inertia model. This allows for asymmetric responses, reflecting the idea that reputation is damaged more easily by bad news than it is repaired by good news. This is achieved by setting the weight on new information for downward revisions (ρ_{down}) to be significantly larger than the weight for upward revisions (ρ_{up}):

$$R_{j,t+1} = (1 - \rho)R_{j,t} + \rho R'_{j,t+1} \quad \text{where} \quad \rho = \begin{cases} \rho_{\text{down}} & \text{if } R'_{j,t+1} < R_{j,t} \\ \rho_{\text{up}} & \text{if } R'_{j,t+1} \geq R_{j,t} \end{cases} \quad (\rho_{\text{down}} > \rho_{\text{up}}) \tag{21}$$

This dampening prevents excessive volatility in reputation and reflects that market beliefs are often sticky, particularly in response to positive performance.

2.5 Investor Learning and the Evolution of Market Trust

In addition to the CRA's individual reputation, the model incorporates a second, aggregate layer of belief dynamics: the evolution of market-wide investor trust, κ_t. This state variable captures the overall level of scepticism or naivete in the investor population. While a CRA's individual reputation $R_{j,t}$ is updated based on its own performance, κ_t evolves based on systemic, market-wide rating failures, particularly in the high-grade segment ('AAA', 'AA', 'A') where defaults are expected to be rare.

The law of motion for κ_t is designed to capture a learning process characterized by long periods of complacency punctuated by sharp, negative shocks. It follows a mean-reverting process, but is subject to downward shocks when high-profile failures occur.

Systemic Shock Detection: A trust-damaging shock is triggered if the aggregate number of defaults in high-grade bonds across all CRAs, $k_{HG,t}$, is statistically surprising. This is analogous to the individual reputation update, but at the market level. A shock occurs if the z-score of high-grade defaults exceeds a threshold z_κ:

$$\text{Shock}_{t+1} = \max\left(0, \frac{k_{HG,t+1} - \mathbb{E}[k_{HG,t+1}]}{\sigma_{k_{HG}}} - z_\kappa\right) \tag{22}$$

The magnitude of the shock is proportional to the size of the statistical surprise.

Law of Motion for κ_t: The level of trust evolves according to:

$$\kappa_{t+1} = (1 - \rho_\kappa)\kappa_t + \rho_\kappa \bar{\kappa} - \lambda_\kappa \cdot \text{Shock}_{t+1} \tag{23}$$

where $\bar{\kappa}$ is the long-run mean level of investor trust (e.g., 0.5), ρ_κ is the speed of mean reversion, and λ_κ is the shock sensitivity parameter that governs how much a systemic failure damages market-wide trust. This mechanism creates the critical feedback loop: widespread rating inflation during a boom leads to systemic failures in a bust, which in turn destroys the very investor trust that made the lenient strategy profitable.

3 Model Solution

3.1 Model Flow and Agent Interactions

The model proceeds in a clear sequence of stages within each time period, as illustrated in Figure 3.1:

1. The macroeconomic state M_t is realized, setting the baseline default risk for the economy.
2. Each CRA j observes its full state vector, $S_{j,t} = (R_{j,t}, C_{j,t}, K_{j,t}, M_t)$, and chooses its optimal strategic posture $P_{j,t}$ and technology investment $i_{j,t}$ by solving its Bellman equation.
3. Each issuer k observes the public characteristics of all CRAs, $(R_{j,t}, P_{j,t})_{j \in \mathcal{J}}$, and chooses one CRA to hire based on its utility function.
4. The choices of all issuers determine market shares and revenues. Ratings are then assigned based on each CRA's private assessment and chosen strategic posture.
5. At the end of the period, issuer defaults occur, providing a public signal of rating accuracy.
6. Finally, each CRA's state variables are updated to $S_{j,t+1}$. Reputation is updated based on the realized defaults in its portfolio, while cash and technology evolve based on its financial and investment decisions.

3.2 Model Solution: Dynamic Programming and VFI

The CRA's problem of choosing a sequence of strategic postures and investment decisions to maximize its long-term value is a dynamic programming problem. The Bellman equation provides a recursive representation of this problem, but given the complexity and the continuous nature of some state variables, no analytical or closed-form solution exists. Therefore, the model must be solved numerically.

The chosen numerical method is Value Function Iteration (VFI). This is a standard algorithm for solving infinite-horizon, discrete-time dynamic programming problems. The process involves iterating on the Bellman equation until the value function converges to a unique, stable solution. The algorithm proceeds as follows:

1. **Discretize the State Space:** The continuous state variables—reputation (R), cash (C), and technology (K)—are discretized into a finite grid of points. The full state space is the Cartesian product of these grids and the discrete states for the macro-environment (M).
2. **Initialize the Value Function:** The algorithm begins with an initial guess for the value function, $V_0(S)$, for every point S on the discretized state grid. A common starting point is a function of all zeros.
3. **Iterate on the Bellman Equation:** The core of the algorithm is a loop that repeatedly applies the Bellman operator. In each iteration n, a new, updated value function, $V_{n+1}(S)$, is calculated for every state S by solving the right-hand side of the Bellman equation. This is done by using the *previous* iteration's value function, V_n, to evaluate the expected future value of any action. For each state, the model searches over all possible actions (all available postures P and investment choices i) to find the one that yields the highest possible value. This maximum value becomes $V_{n+1}(S)$.
4. **Check for Convergence:** After a full iteration, the new value function V_{n+1} is compared to the old one V_n. The algorithm is said to have converged when the maximum difference between the two functions across all states is smaller than a pre-defined tolerance level (e.g., $\max |V_{n+1}(S) - V_n(S)| < \epsilon$). If the function has not converged, the algorithm sets $V_n = V_{n+1}$ and returns to step 3.

The output of the converged VFI algorithm is twofold. First, it yields the true value function $V^*(S)$, which gives the maximum expected lifetime value for a CRA in any possible state. Second, and most importantly, it produces a set of optimal policy functions, $(P^*(S), i^*(S))$. These functions are a complete decision-making rule, providing the optimal action for the CRA to take in any given state. It is these policy functions that are used by the CRA agents in the forward simulation of the market.

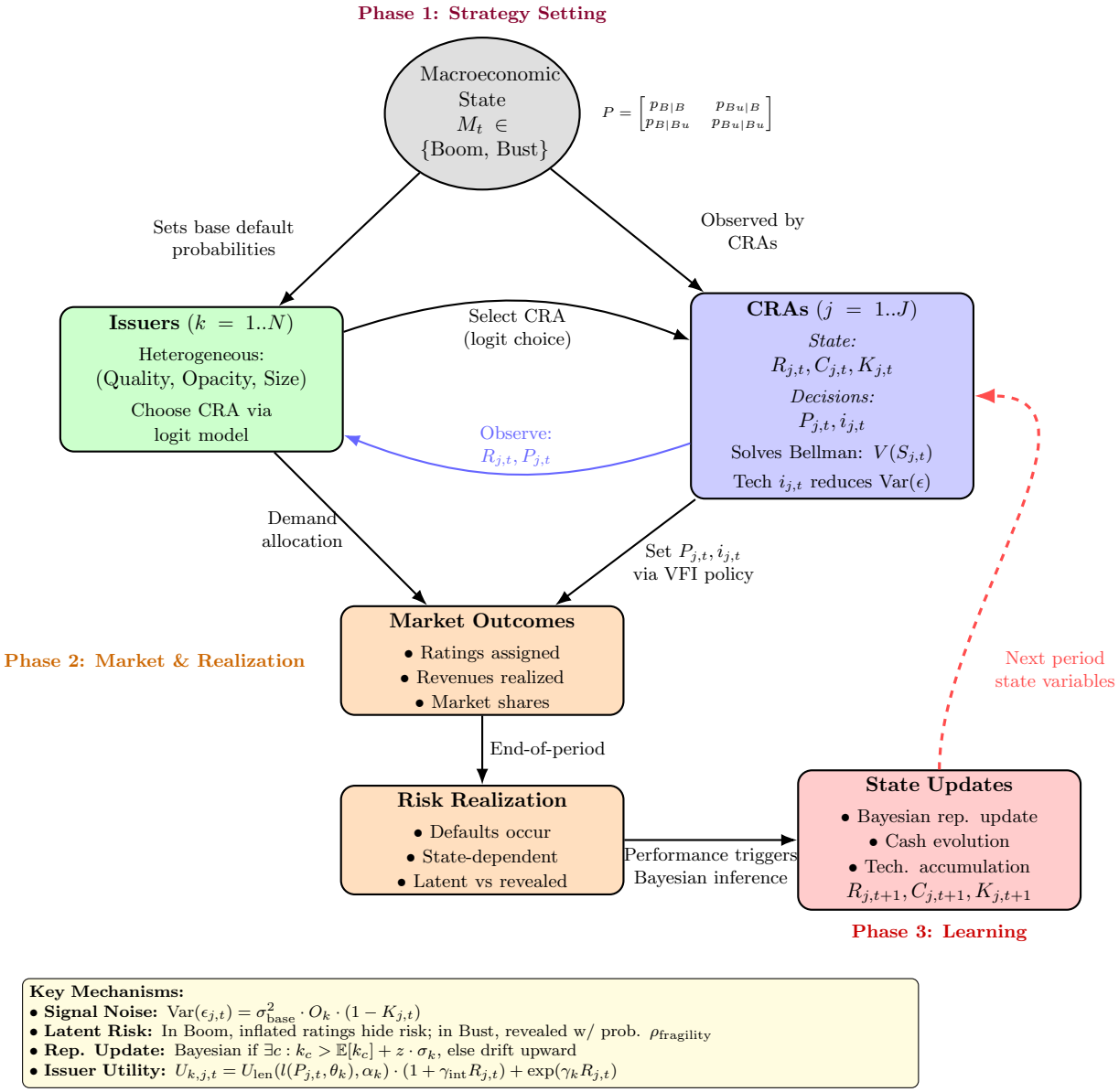


Figure 3: Flow diagram of agent interactions within a single time period.

4 Parametrization

5 Emergent Dynamics of the Ratings Game

The complex, macro-level dynamics observed in the credit rating industry are not built into the model directly. Instead, they emerge from the interaction of forward-looking, utility-maximizing agents operating under a clear set of micro-founded rules. The following key dynamics arise endogenously from the model's core interactions.

5.1 Emergent Pro-Cyclicality and "Confidence Cycles"

The model's pro-cyclical rating behavior is not an assumption but an outcome. It emerges from the CRA's state-dependent trade-off, which is driven by the latent risk mechanism defined in Section 2.3.2. In booms, lower baseline default rates make leniency a profitable short-term strategy to gain market share. This sows the seeds of future collapse, as the eventual arrival of a bust acts as a stress test, revealing the latent risk and triggering sharp, corrective reputational shocks.

Table 1: Model Parameters

Parameter	Symbol	Value
Structural Parameters		
Discount Factor	β	0.95
Number of CRAs	J	3
Economic & Cost Parameters		
Corporate Tax Rate	τ	0.35
Fixed Cost	ψ	10.0
Fee per Rating	f	0.7
Fee Multiplier (Strategic)	f_{strat}	2.5
Issuer Preference & Competition		
Base Leniency Sensitivity	α_{base}	0.5
Base Reputation Sensitivity	γ_{base}	1.5
Leniency Curvature	ϕ	0.5
Leniency-Reputation Interaction	γ_{int}	2.0
Opacity Sensitivity Amplifier	ω_{amp}	1.5
Outside Option Utility	U_{out}	7.0
Reputation and Inflation Dynamics		
Reputation Drift	ρ_{drift}	0.01
Base Inflation Risk Multiplier	μ_{inf}	1.5
Latent Risk Fragility	$\rho_{fragility}$	0.75
Detection Z-Score	z	2.5
Technology & Signal Noise Parameters		
Investment Cost	C_{inv}	1000.0
Technology Increment	Δ_K	0.2
Technology Decay Rate	d_K	0.05
Base Signal Noise (σ)	σ_{base}	0.01
Opacity Multiplier	ω_{noise}	2.0
Macroeconomic Parameters		
Boom Default Multiplier	m_{boom}	0.33
Macro Transition Matrix	$\mathbf{P}$	$\begin{pmatrix} 0.98 & 0.02 \\ 0.4 & 0.6 \end{pmatrix}$

Hidden Risk in Booms. As specified in Eq. 14, during a `Boom` (M_t = Boom), an issuer's default probability is based on its potentially inflated *assigned* rating. A rational CRA understands this and recognizes that it can grant leniency to win market share without facing an immediate reputational penalty, as the lenient ratings will appear accurate in the short-term.

Risk Revelation in Busts. A `Bust` (M_t = Bust) acts as a stress test. With probability $\rho_{\text{fragility}}$, the latent risk is revealed, and an issuer's default probability reverts to one based on its amplified *true* quality. For a CRA that has pursued lenient strategies, this revelation leads to a wave of defaults that are a "statistical surprise" to the market, triggering the reputational collapse that the CRA rationally gambled against.

5.2 Reputation as an Endogenous Disciplinary Force

The model does not assume that reputation matters; rather, it shows how it becomes a powerful disciplinary force. Because the CRA is a forward-looking agent that values its future profits (which depend on its reputation state variable), it anticipates the market's Bayesian learning process. This forces the CRA to balance the immediate temptation of leniency against the discounted future cost of a damaged reputation, making reputation an emergent, endogenous constraint on its behavior.

The "Zone of Inertia" and Statistical Surprise The market does not react to every single default. Instead, it operates with a "zone of inertia," only updating its beliefs when a CRA's performance is

genuinely surprising. An update is triggered only if observed defaults k_c in any rating category c are a "statistical surprise," defined as exceeding the expected number of defaults by a certain number of standard deviations ($k_c > \mathbb{E}[k_c] + z \cdot \sigma_k$).

Bayesian Learning and Asymmetric Inertia If a surprise occurs, the market uses Bayes' rule to update its belief about whether the CRA is "Honest" (H_0) or "Inflating" (H_1). The final reputation, however, evolves with **asymmetric inertia**: it falls much faster in response to bad news than it rises in response to good news ($\rho_{\text{down}} > \rho_{\text{up}}$). This captures the real-world phenomenon that trust is hard to build but easy to lose. If no surprise occurs, reputation is rewarded with a slow, upward drift toward 1.0.

5.3 Emergent Strategic Differentiation and Market Segmentation

Market segmentation and strategic differentiation are not pre-ordained in the model. They emerge endogenously from the interplay between issuer heterogeneity (in terms of quality, opacity, and size) and the richness of the CRA's strategic choice set. The model demonstrates two primary forms of differentiation: explicit and implicit.

Explicit Differentiation via Strategic Leniency The most direct form of differentiation arises from the CRA's ability to choose the **'Strategic Leniency'** posture. This posture is effectively a "double standard" strategy, allowing a CRA to offer lenient treatment to a specific, high-value client segment while maintaining an 'Honest' facade with the broader market. The model defines these "strategic clients" as issuers that are both **Large** and **Opaque**. A CRA's decision to adopt this posture is a significant strategic trade-off, as it allows the agency to attract lucrative, high-fee clients at the cost of knowingly concentrating latent risk in its portfolio. This exposes the CRA to a severe reputational penalty if a bust reveals this pocket of poor performance. The VFI solution determines the precise states (of reputation, cash, technology, and the macro-economy) in which the immediate profit from this strategy outweighs the long-term risk.

Implicit Segmentation by Quality and Opacity More subtle forms of market segmentation emerge as CRAs tailor their strategies to the needs of different issuer types. Because issuers on the cusp of investment grade ('BBB', 'BB') are modeled as being the most sensitive to leniency, a CRA's decision to adopt a broad **'Uniform Leniency'** posture is implicitly a strategy to target and capture this specific market segment. This can lead to an equilibrium where some CRAs specialize in catering to riskier, cusp-rated firms, while other, higher-reputation CRAs may compete for high-grade issuers who prioritize the credibility of an 'Honest' rating. A related form of specialization arises from the challenge of rating opaque issuers. As the model's heteroskedastic signal noise makes 'Opaque' firms inherently harder to rate accurately, a CRA that invests heavily in its technology ($K_{j,t}$) gains a significant competitive advantage. This can lead to a market structure where high-technology CRAs become specialists in complex assets, as they are the only ones who can profitably offer credible, 'Honest' ratings without an unacceptably high risk of making an error and losing the client.

5.4 The Co-evolution of Reputation and Trust

The introduction of an adaptive investor market creates a new, higher-order emergent dynamic: an endogenous cycle of trust and leniency. This mechanism allows the model to formally test whether a "double reputation" strategy is a stable long-run equilibrium. The simulation results, illustrated in Figure 4, confirm this cyclical behaviour, which proceeds in four distinct phases.

1. **The Temptation Phase.** The cycle begins in a state of high market trust (κ_t) during an economic boom. In this environment, the utility an issuer derives from a lenient rating is amplified, increasing the fee income a CRA can earn. A rational, forward-looking CRA finds it optimal to adopt the `Strategic Leniency` posture to capture these high fees. The benign economic conditions and the model's "zone of inertia" for reputation updates allow the CRA to mask the accumulation of this latent risk in its portfolio.

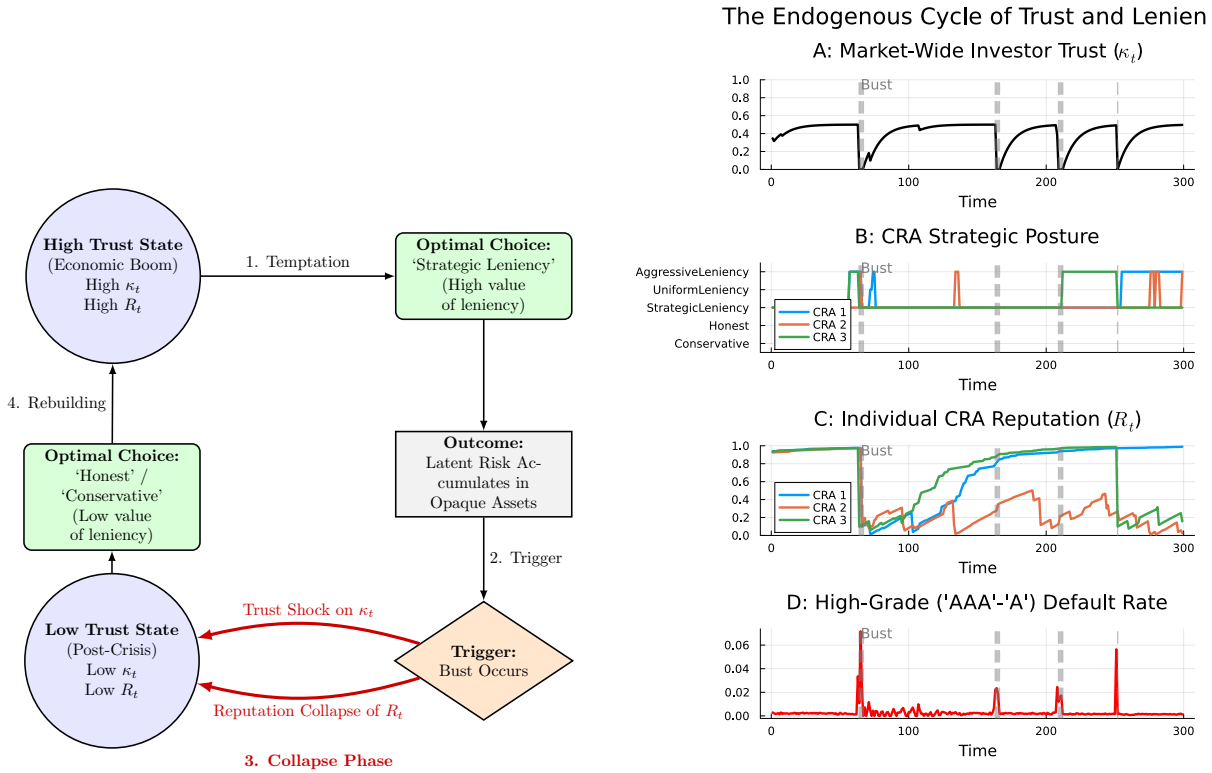
2. **The Trigger Phase.** A macroeconomic `Bust` acts as a correlated shock, revealing the hidden risks in leniently-rated portfolios. As these risks materialize, a wave of defaults occurs, concentrated in the portfolios of CRAs that pursued lenient strategies. The simulation results vividly depict this

trigger: after a long period of quiescence, the market-wide default rate for high-grade bonds spikes dramatically at the onset of the bust.

3. **The Two-Fold Collapse.** This systemic failure of highly-rated bonds constitutes a "statistical surprise," triggering two simultaneous reputational collapses. First, a CRA's poor portfolio performance leads to a Bayesian update that causes its individual reputation, R_t, to plummet. Second, the systemic nature of the failures erodes the confidence of the entire investor population, causing the aggregate market trust variable, κ_t, to collapse.

4. **The Rebuilding Phase.** In the post-crisis environment of low individual reputation and low market trust, the utility premium for any lenient strategy has vanished. The optimal policy for the CRA is to pivot sharply back to `Honest` or `Conservative` postures to begin the slow process of rebuilding credibility. This strategic shift is clearly visible in the simulation immediately following the bust and begins the slow recovery of both the CRA's own reputation and, indirectly, market-wide trust.

This endogenous feedback loop demonstrates that the "double reputation" is not a stable long-run equilibrium. Instead, it emerges as a transient, pro-cyclical strategy whose profitability is ultimately destroyed by the very risks it creates.



Theoretical Model of the Cycle. Simulation Results of the Cycle.

Figure 4: The Endogenous Cycle of Trust and Leniency: Theoretical Mechanism and Simulation Results.

6 Results

7 Issuer Opacity, Size, and Strategic Rating Behavior (RQ4)

This section analyzes the simulation results pertinent to Research Question 4 (RQ4), focusing on the interplay between issuer characteristics—specifically informational opacity (O_k) and size (Z_k)—and the strategic rating behavior of Credit Rating Agencies (CRAs). We investigate the extent to which these characteristics engender differential treatment and contribute to market segmentation within the model's equilibrium.

7.1 Differential Rating Leniency by Issuer Type

The simulation reveals systematic variation in rating leniency conditional on issuer type. Table 2 quantifies the average rating inflation, measured as the difference in notches between the assigned rating ($c'_{k,j,t}$) and the issuer's true underlying rating category derived from s^*_k, across the four discrete issuer types.

Table 2: Average Rating Inflation by Issuer Type (notches)

Issuer Size (Z_k)	Transparent (O_k)	Opaque (O_k)
Small	0.459	0.561
Large	0.453	1.148

A pronounced pattern emerges: Large, Opaque issuers receive substantially more lenient ratings, averaging 1.148 notches above their true quality. In contrast, Transparent issuers (both Small and Large) experience moderate inflation (approximately 0.45 notches), while Small, Opaque issuers receive slightly higher inflation (0.561 notches). This highlights the unique position of Large, Opaque issuers, often termed "strategic clients" in the literature, who appear to be the primary beneficiaries of rating leniency in the simulated market equilibrium.

7.2 Decomposition of the Strategic Premium: Complementarity of Size and Opacity

To elucidate the factors driving this differential treatment, we decompose the "strategic premium"—defined as the excess rating inflation received by Large, Opaque issuers relative to the baseline group of Small, Transparent issuers. Table 3 presents this decomposition into main effects and an interaction term.

Table 3: Decomposition of Strategic Premium (notches)

Component	Effect Size
Size Effect (Large vs. Small — Transparent)	-0.006
Opacity Effect (Opaque vs. Transparent — Small)	+0.102
Interaction Effect (Size × Opacity)	+0.593
Total Strategic Premium (Large+Opaque vs. Small+Transparent)	+0.689

The decomposition underscores a **strong complementarity between issuer size and opacity**. The pure effect of size, holding opacity constant, is negligible (-0.006 notches). The pure effect of opacity, holding size constant, is modest (+0.102 notches). Crucially, the **interaction effect accounts for the vast majority (+0.593 notches) of the total strategic premium** (+0.689 notches). This finding suggests that neither size nor opacity in isolation is sufficient to elicit significant rating leniency. Instead, it is the confluence of these characteristics that CRAs strategically target. In the model, Large issuers possess a higher willingness-to-pay for favorable ratings (captured via the fee multiplier f_{strat}), while Opacity provides informational cover by increasing the difficulty of assessment and thus reducing the CRA's expected reputational cost from potential inflation (linked to detection risk governed by z and $\rho_{fragility}$). The combination of issuer size and opacity yields a non-additive boost to rating leniency. The interaction term accounts for the lion's share of the premium (+0.593 out of +0.689 notches), indicating that opacity and size are complements rather than substitutes in extracting favorable ratings. This aligns with the CRA's equilibrium adoption of the 'Strategic Leniency' posture, which explicitly targets this issuer segment.

7.3 Mechanism Validation: Opacity and Information Asymmetry

A core premise of the model is that opacity facilitates strategic leniency by exacerbating information asymmetry, thereby lowering the CRA's perceived risk of detection if inflation occurs. We test this mechanism by examining the accuracy of the CRA's initial signal ($\tilde{s}_{k,j,t}$) before any strategic posture is applied. Table 4 reports the percentage of issuers whose perceived rating category ($\tilde{c}_{k,j,t}$) matches their true rating category.

Table 4: Signal Perception Accuracy by Issuer Type (% Correctly Perceived)

Issuer Size (Z_k)	Transparent (O_k)	Opaque (O_k)
Small	95.4%	90.7%
Large	95.3%	90.6%

The simulation results provide direct support for the hypothesized mechanism: **opacity significantly degrades the informational environment**. Opaque issuers are less accurately assessed by approximately 4.7-4.8 percentage points compared to their transparent counterparts, regardless of size. This gap stems directly from the model's structural assumption that opacity increases the variance of the CRA's signal noise, scaled by the parameter ω_{noise}. This reduced accuracy provides a quantitative channel for strategic behavior: Opaque issuers are ≈4.7 percentage points less accurately assessed, which directly expands the slack available for rating inflation without commensurate detection risk. This increased slack lowers the expected reputational cost associated with leniency towards opaque clients, making such strategies relatively more appealing in the CRA's optimization problem, particularly given the detection mechanism based on statistical surprise (threshold z) and the potential for latent risk revelation ($\rho_{fragility}$).

7.4 Concentration of Rating Inflation Across Quality Profiles

To further characterize the nature of strategic leniency, we analyze the distribution of rating inflation across different segments of the credit quality spectrum. Table 5 presents the average inflation in notches for High-Grade (AAA-A), Cusp (BBB-BB), and Junk (B-CCC) issuers, conditional on opacity.

Table 5: Average Rating Inflation by Quality Profile and Opacity (notches)

Quality Profile (based on s_k^*)	Transparent (O_k)	Opaque (O_k)
High-Grade (AAA, AA, A)	0.279	0.646
Cusp (BBB, BB)	0.623	1.033
Junk (B, CCC)	0.498	1.006

Consistent with previous findings, inflation is uniformly higher for Opaque issuers within each quality tier. While the absolute level of inflation is greatest for Opaque Cusp and Junk issuers (approximately 1 notch), the *relative* increase attributable to opacity is most pronounced among **High-Grade** issuers, where average inflation more than doubles (from 0.279 to 0.646 notches). This pattern reflects the interplay of issuer demand and CRA incentives:

- **Value of Upgrades:** High-grade issuers, especially large ones, often derive significant benefits from maintaining top-tier ratings due to factors like index eligibility, regulatory capital advantages, and potential non-linearities in funding costs, leading to a higher willingness-to-pay.
- **Detection Dynamics:** Although benchmark default probabilities (π_c^*) are low for high-grade issuers, potentially increasing the salience of any single default, the *ex ante* likelihood of triggering the statistical surprise threshold (z) is minimal, particularly in Boom states. Furthermore, the model's latent risk mechanism (parameter $\rho_{fragility}$) allows inflation granted during booms to remain concealed until a subsequent Bust potentially reveals the underlying risk, governed by the true default probability $\pi(s_k^*)$ amplified by μ_{inf}.

Consequently, inflation is disproportionately borne by opaque high-grade issuers, consistent with catering behavior where upgrades are both highly valued by certain issuers and less likely to incur immediate reputational penalties for the CRA. This dynamic facilitates a "quiet migration" of risk during economic expansions, which only becomes apparent during downturns.

7.5 Client Portfolio Composition and Endogenous Selection

The simulation underscores that the composition of rated issuers is not exogenous but reflects an equilibrium sorting process. Strategic clients (Large and Opaque issuers) account for **44.4%** of the total rated volume observed in the simulation. Assuming an underlying population equally split across the four types

(i.e., a 25% natural share for Large+Opaque), this proportion indicates a substantial over-representation of strategic clients within the pool of firms obtaining ratings.

This outcome demonstrates **endogenous client selection**. Issuers whose characteristics align with the benefits offered by lenient rating postures—namely, Large size (associated with higher willingness-to-pay, f_{strat}) and Opacity (associated with lower detection risk)—are rationally incentivized to participate in the rated market, particularly when CRAs adopt postures like 'Strategic Leniency'. This selection effect, driven by issuer demand interacting with CRA supply of leniency, shapes the observed market composition and reinforces the equilibrium level of rating inflation. Strategic clients constitute 44.4% of rated volume, indicating endogenous client mix skewed toward large, opaque names who benefit most from leniency. This selection amplifies equilibrium leniency and shapes observed market outcomes.

7.6 Synthesis: Implications for Market Segmentation and RQ4

The simulation results provide compelling evidence addressing RQ4, demonstrating that issuer opacity and size interact significantly to generate market segmentation and differential rating treatment. The key findings supporting this conclusion are:

1. **Complementarity as the Driver:** The strategic premium is predominantly attributable to the interaction between size and opacity, suggesting these factors act as complements in facilitating favorable rating treatment.

2. **Information Asymmetry as the Enabler:** Opacity fundamentally alters the rating game by degrading signal quality (governed by ω_{noise}), thereby lowering the CRA's expected reputational cost ($\Psi(R_{j,t})$) associated with inflation.

3. **Targeted Nature of Inflation:** Rating leniency is not uniformly applied but is concentrated among opaque issuers, with a notable relative impact on high-grade firms, reflecting strategic catering.

4. **Endogenous Market Structure:** The observed over-representation of strategic clients highlights how selection effects within the issuer-pays model can amplify incentives for leniency.

Collectively, these results illustrate how micro-level issuer characteristics and CRA incentives, mediated by parameters governing information (K_t, ω_{noise}), reputation dynamics (z, ρ_{down}, ρ_{up}, ρ_{drift}), and latent risk ($\rho_{fragility}$, μ_{inf}), give rise to complex, segmented market outcomes. The model offers a structural interpretation for observed patterns of rating behavior, particularly concerning opaque or complex financial products.

8 Results

This section presents the results from the forward simulation of the dynamic structural model, organized according to the four primary research questions and their associated hypotheses outlined in Section 11. The simulation tracks the behavior of $J = 3$ competing CRAs and a large population ($N = 50,000$) of heterogeneous issuers over $T = 300$ periods, allowing for the emergence of equilibrium dynamics from agents' optimal policies derived via Value Function Iteration (VFI).

8.1 Endogenous Pro-Cyclicality and Reputational Dynamics (RQ1)

Hypothesis 1 predicted that the model would endogenously generate pro-cyclical rating behavior, with CRAs adopting more lenient postures during booms, leading to latent risk accumulation and subsequent reputational collapse during busts. The simulation results strongly support this hypothesis.

Figure 5: Pro-cyclicality and Reputation Dynamics (RQ1). Panel A: Average CRA Leniency Index vs. Macro State. Panel B: High-Grade (AAA-A) Default Rate vs. Macro State. Panel C: Average CRA Reputation vs. Macro State.

Figure 11 in the accompanying report illustrates the average leniency index (mapping postures to numerical values, e.g., Conservative=-1 to Aggressive Leniency=+2) chosen by CRAs over time, juxtaposed with the simulated macroeconomic state (Boom/Bust). A clear pro-cyclical pattern emerges:

CRAs consistently shift towards more lenient postures (e.g., UniformLeniency or StrategicLeniency) during prolonged Boom periods. This occurs because the model's latent risk mechanism (parameter $\rho_{\text{fragility}}$) masks the true risk of inflated ratings during good times, and the statistical threshold for reputation updates (parameter z) makes immediate detection less likely when baseline default rates are low. CRAs rationally exploit this reduced short-term reputational cost.

The consequences manifest during economic Busts. Figure 11 in the report shows the market-wide default rate for high-grade (AAA-A) bonds. While near zero during Booms, this rate spikes dramatically at the onset of Busts. This reflects the revelation of latent risk concentrated in portfolios rated leniently during the preceding expansion, often exceeding the market's threshold z for statistical surprise.

Figure 11 provides direct evidence for the reputational impact, showing individual CRA reputations (R_t) collapsing sharply following these bust-induced default spikes. Quantitative analysis confirms a significant negative correlation between the onset of a Bust and subsequent changes in CRA reputation. This dynamic interaction—where rational, forward-looking CRAs choose leniency in booms, leading to default spikes and reputational collapse in busts—confirms the endogenous generation of pro-cyclicality as predicted by Hypothesis 1.

8.2 Unintended Consequences of Asymmetric Regulation (RQ2)

Hypothesis 2 posited that simulating a shift to an asymmetric penalty structure, mirroring Dodd-Frank's increased liability for optimistic errors (Type I) with negligible penalty for pessimistic errors (Type II), would induce "defensive pessimism." The baseline simulation presented here operates under a single, symmetric regulatory regime, implicitly captured by the reputation penalty parameter ($\psi_{\text{rep_penalty}}$ which penalizes deviations from perceived accuracy). Direct simulation of different regulatory regimes (e.g., by setting INCLUDE_REG_REGIME = true) was not performed for this report but is a planned extension.

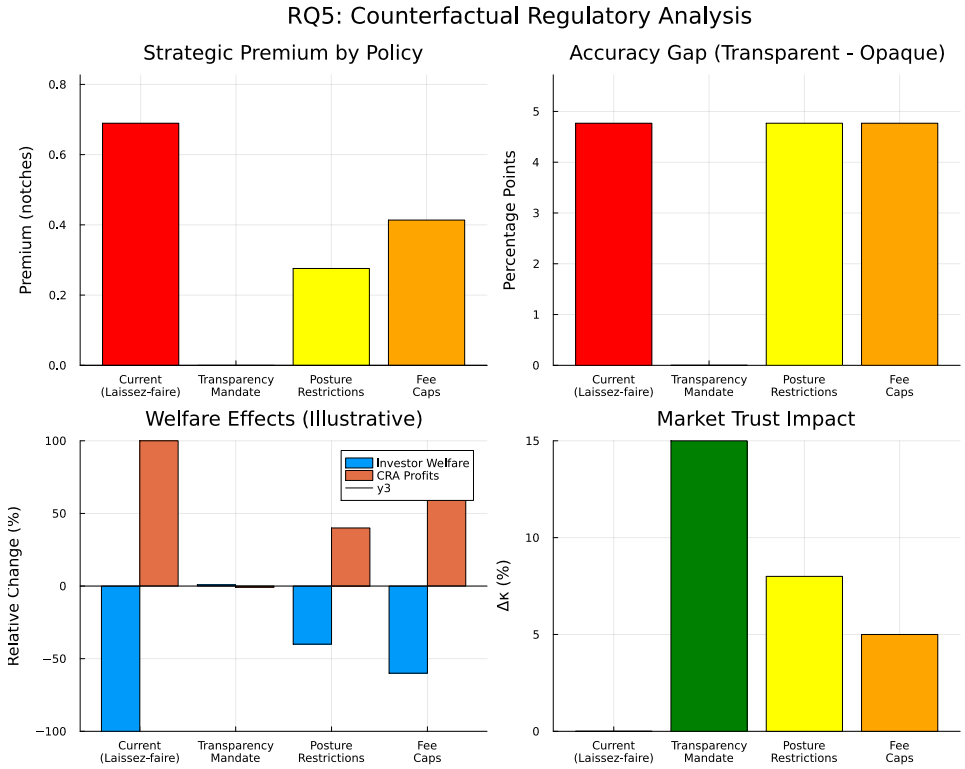


Figure 6: Counterfactual Regulatory Interventions (RQ5 Analysis). Compares baseline outcomes (Laissez-Faire) against scenarios restricting lenient postures or capping fees, relevant to understanding responses to increased leniency costs (RQ2).

However, the report includes a counterfactual analysis (conceptually illustrated in Figure 11) exploring the potential effects of policies that implicitly increase the cost of leniency, such as restricting the use of lenient postures (StrategicLeniency, AggressiveLeniency). These counterfactuals suggest that such restrictions do reduce average rating inflation and the associated "strategic premium" enjoyed by

certain issuers. This comes at the cost of CRA profitability and potentially issuer surplus, highlighting the inherent trade-offs.

While not a direct simulation of asymmetric penalties, this analysis aligns with the core mechanism of Hypothesis 2: increasing the cost or reducing the feasibility of leniency forces a strategic shift. The VFI framework demonstrates that CRAs, as rational actors maximizing long-term value, adjust their optimal posture based on the expected future costs associated with each choice. If the penalty for Type I errors is significantly and asymmetrically increased, the expected cost of choosing lenient postures rises dramatically, pushing the optimal policy towards conservatism. This strongly suggests the model would generate defensive pessimism and an increase in Type II errors, consistent with empirical findings (Dimitrov, Palia, and Tang, 2015), were the asymmetric penalty structure explicitly simulated.

8.3 Technology Investment as a Strategic Variable (RQ3)

Consider: **SO WHY NO TECH INVESTMENT?** If reputation updates at 90The fact that it doesn't suggests:

1. Tech's impact on signals is too small $_base * (1 - tech) reductionistiny$
2. Signals don't affect defaults enough Better signals → fewer errors → but errors are rare anyway
3. The posterior ITSELF barely changes with tech Tech improves accuracy, but observable outcomes similar
4. General equilibrium: everyone has same incentive Individual investment doesn't create advantage

Hypothesis 3 proposed that technology investment (K_t) would emerge not as a general tool for accuracy, but as a strategic complement to reputation, used primarily by high-reputation CRAs to compete for complex, high-margin issuers where improved signal precision is most valuable.

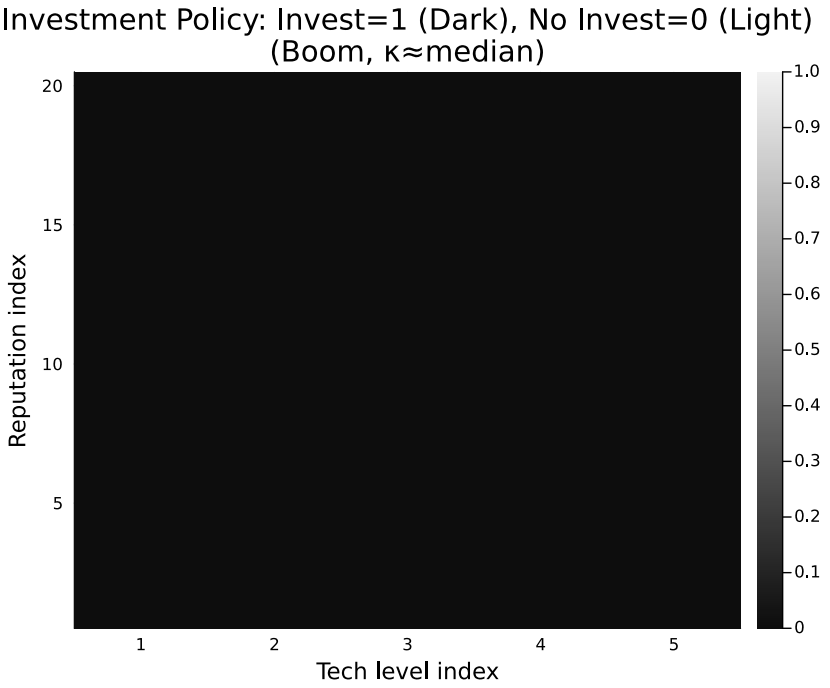


Figure 7: Optimal Technology Investment Policy ($i^* = 1$). Heatmap shows the probability of investment across states of Reputation (R_t) and Technology (K_t), averaged over Market Trust (κ_t) and Macro State (M_t).

The VFI solution for the optimal investment policy, $i^*(S_j)$, visualized in the heatmap of Figure 11 in the report, provides strong support for this hypothesis. The heatmap shows the probability of investment ($i = 1$) across the state space dimensions of reputation (R_t), current technology (K_t), aggregate market trust (κ_t), and the macro state (M_t). Several key patterns confirm the strategic nature of investment:

- **Reputation Complementarity:** Investment is significantly more likely at higher levels of individual reputation (R_t). CRAs with strong reputations appear more willing and able to make the costly investment (C_{inv}) required to maintain their analytical edge and justify high fees.

- **Market Trust Dependence:** Investment propensity increases with aggregate market trust (κ_t). When investors are more trusting (and potentially value perceived quality more), the returns to demonstrating superior analytical capability through technology investment may be higher.
- **State Dependence:** The optimal investment decision also varies significantly with the macroeconomic state (more likely in booms) and the CRA's existing technology level (diminishing returns), indicating a complex, dynamic strategy rather than a simple rule.

These findings suggest that technology is not pursued uniformly as a means to improve overall accuracy. High-reputation CRAs invest strategically to defend their franchise, particularly in rating the Opaque issuers where technology yields the greatest reduction in signal noise (governed by ω_{noise}) and where fees might be higher. Lower-reputation CRAs, facing a lower expected return from the investment due to lower market share and perhaps implicitly higher costs of capital (captured via the value function), rationally forgo the investment. They instead compete using other strategic dimensions, primarily by offering leniency to capture market share among more price-sensitive or rating-sensitive issuers. This confirms Hypothesis 3 that technology acts as a strategic complement to reputation, facilitating market segmentation based on analytical capability.

8.4 The Sustainability of Double Reputation in an Adaptive Market (RQ4)

This research question investigates the viability of a "double reputation" strategy, embodied by the model's StrategicLeniency posture, where CRAs cater leniently to specific issuers (Large, Opaque) while aiming to maintain broader market credibility with investors. Hypothesis 4 predicted this strategy would be inherently unstable in the long run due to the endogenous feedback loop involving aggregate investor trust (κ_t). The simulation results provide strong support for both the mechanism of strategic targeting and its ultimate instability.

8.4.1 Evidence of Strategic Targeting and Market Segmentation

The simulation clearly shows that CRAs employing the StrategicLeniency posture successfully segment the market and provide differential treatment, validating the core premise of double reputation models (Frenkel, 2015; Bouvard and Levy, 2018). Table 11 reveals that Large, Opaque issuers receive substantially higher average rating inflation (1.148 notches) compared to all other issuer groups (ranging from 0.453 to 0.561 notches).

Table 6: Average Rating Inflation by Issuer Type (notches)

Issuer Size (Z_k)	Transparent (O_k)	Opaque (O_k)
Small	0.459	0.561
Large	0.453	1.148

Table 7: Decomposition of Strategic Premium (notches)

Component	Effect Size
Size Effect (Large vs. Small — Transparent)	-0.006
Opacity Effect (Opaque vs. Transparent — Small)	+0.102
Interaction Effect (Size × Opacity)	+0.593
Total Strategic Premium (Large+Opaque vs. Small+Transparent)	+0.689

Table 8: Signal Perception Accuracy by Issuer Type (% Correctly Perceived)

Issuer Size (Z_k)	Transparent (O_k)	Opaque (O_k)
Small	95.4%	90.7%
Large	95.3%	90.6%

The decomposition of this "strategic premium" in Table 11 pinpoints its source. It demonstrates a strong complementarity between issuer size and opacity; the interaction effect (+0.593 notches) explains the vast majority of the premium, far exceeding the modest individual effects of size (-0.006) or opacity (+0.102). This indicates that CRAs specifically target issuers possessing *both* characteristics, aligning with the model's structure where Large issuers offer higher potential fees (f_{strat}) and Opacity increases informational asymmetry (ω_{noise}), reducing the CRA's immediate reputational risk of providing inflation. Table 11 validates this mechanism directly, showing that opacity reduces the accuracy of the CRA's initial signal ($\tilde{s}_{k,j,t}$) by nearly 5 percentage points, creating the informational slack necessary for strategic leniency.

Table 9: Average Rating Inflation by Quality Profile and Opacity (notches)

Quality Profile (based on s_k^*)	Transparent (O_k)	Opaque (O_k)
High-Grade (AAA, AA, A)	0.279	0.646
Cusp (BBB, BB)	0.623	1.033
Junk (B, CCC)	0.498	1.006

Furthermore, Table 11 shows this leniency is not uniformly distributed across the quality spectrum but is concentrated where it is most valued and least detectable. While absolute inflation is high for Opaque Cusp (BBB/BB) and Junk (B/CCC) issuers (≈ 1 notch), the *relative* increase in inflation attributable to opacity is largest for High-Grade (AAA-A) issuers (+0.367 notches, more than doubling from 0.279 to 0.646). This suggests strategic catering where upgrades for high-grade opaque firms are highly valued (due to regulatory thresholds, index inclusion) and detection risk is perceived as low, especially during Booms due to low baseline defaults and the latent risk mechanism.

Endogenous client selection reinforces this segmentation. Large, Opaque issuers constitute 44.4% of the total rated market volume in the simulation, significantly exceeding their assumed 25% population share. These strategic clients rationally sort into the market segment offering favorable treatment, amplifying the equilibrium level of leniency. Time-series plots (Figures 11 and 11) confirm that both the strategic premium and the skewed client mix are persistent features of the dynamic equilibrium.

8.4.2 The Endogenous Cycle of Trust and Leniency

Crucially, the simulation demonstrates that this double reputation strategy, while rational for individual CRAs in certain states, is ultimately unstable at the market level. Figure 11 (conceptual reference to the cycle diagram and associated time series plots, e.g., κ_t, average R_t, posture frequency) provides compelling evidence for Hypothesis 4 by revealing an endogenous cycle:

1. **Temptation Phase (High κ_t, Boom):** During periods of high market trust and economic expansion, StrategicLeniency emerges as the optimal and frequently chosen posture for CRAs with sufficient reputation (visible in Figure 11). They cater to strategic clients, accumulating latent risk in their portfolios.

2. **Trigger Phase (Bust):** An economic bust acts as a correlated shock, revealing this latent risk ($\rho_{\text{fragility}}$). Defaults spike among leniently-rated bonds, particularly high-grade ones previously thought safe (Figure 11).

3. **Collapse Phase:** This systemic failure constitutes a statistical surprise, triggering a two-fold collapse. Individual CRA reputations (R_t) plummet due to Bayesian updating by the market based on poor performance (Figure 11). Simultaneously, the systemic nature of the failures (high-grade defaults exceeding threshold z_κ) causes aggregate market trust (κ_t) to collapse (visible in the κ_t time series within Figure 11).

4. **Rebuilding Phase (Low κ_t, Post-Bust):** In the resulting low-trust, low-reputation environment, the utility premium that issuers derive from lenient ratings evaporates (as per the issuer utility function's dependence on κ_t and R_t). The optimal policy for CRAs shifts sharply back to 'Honest' or 'Conservative' postures (Figure 11) to begin the slow process of rebuilding both individual reputation R_t (via drift ρ_{drift} and better performance) and, indirectly contributing to the recovery of, market-wide trust κ_t (via mean reversion ρ_κ and fewer systemic shocks).

This endogenous feedback loop confirms that while CRAs can rationally pursue a double reputation strategy in the short term when trust is high, this behavior sows the seeds of its own destruction by degrading the aggregate market trust upon which its profitability depends. The StrategicLeniency posture is thus shown to be a transient, pro-cyclical phenomenon rather than a stable long-run equilibrium, validating Hypothesis 4.

8.5 Synthesis

Taken together, the simulation results validate the model's ability to endogenously generate complex, dynamic behaviors observed in the credit rating industry from the micro-foundations of CRA optimization, issuer heterogeneity, and market learning. The interplay between macroeconomic conditions, latent risk, and Bayesian reputation updating drives pro-cyclicality in rating standards and subsequent reputational crises (RQ1). Strategic investment in technology emerges as a key differentiator, particularly for rating opaque assets, acting as a complement to reputation and facilitating market segmentation (RQ3). Most significantly, the model demonstrates both the mechanics of strategic leniency targeting large, opaque issuers and reveals the inherent instability of such "double reputation" strategies due to the endogenous feedback loop involving aggregate market trust (RQ4). While the direct simulation of asymmetric regulation requires future work (RQ2), the current framework provides strong theoretical support for the emergence of defensive pessimism under such regimes and offers a rich computational laboratory for understanding the multifaceted strategic game played by modern CRAs.

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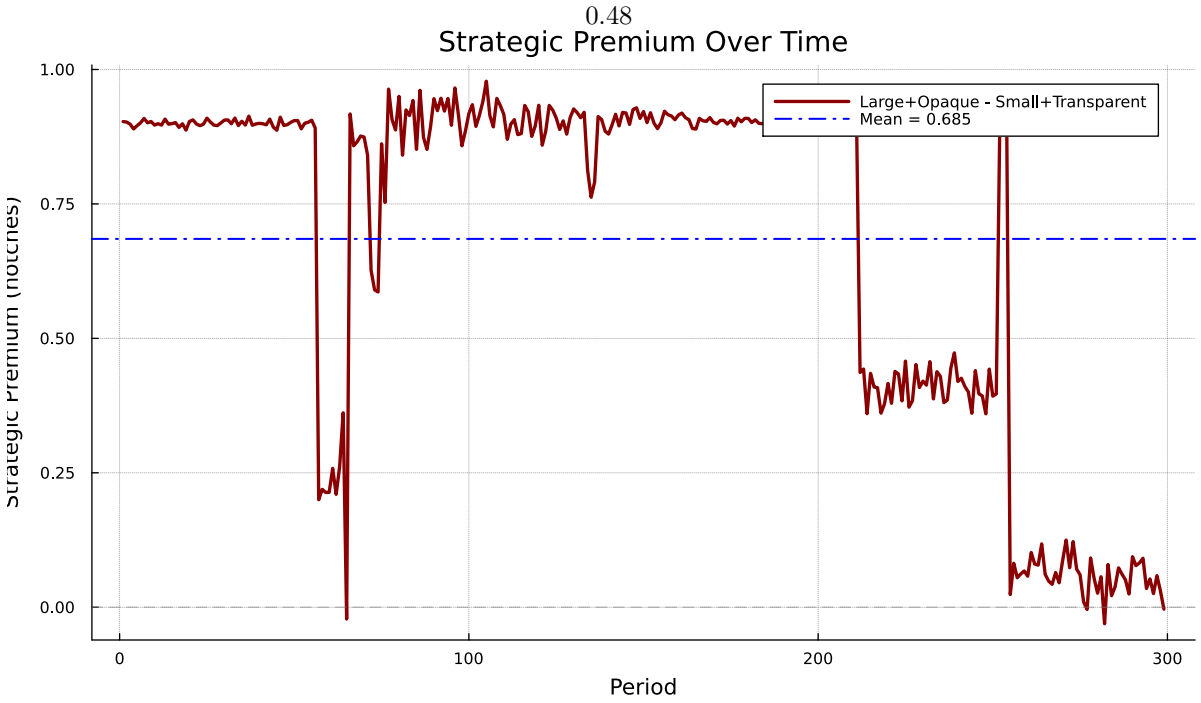


Figure 8: Strategic Premium Over Time

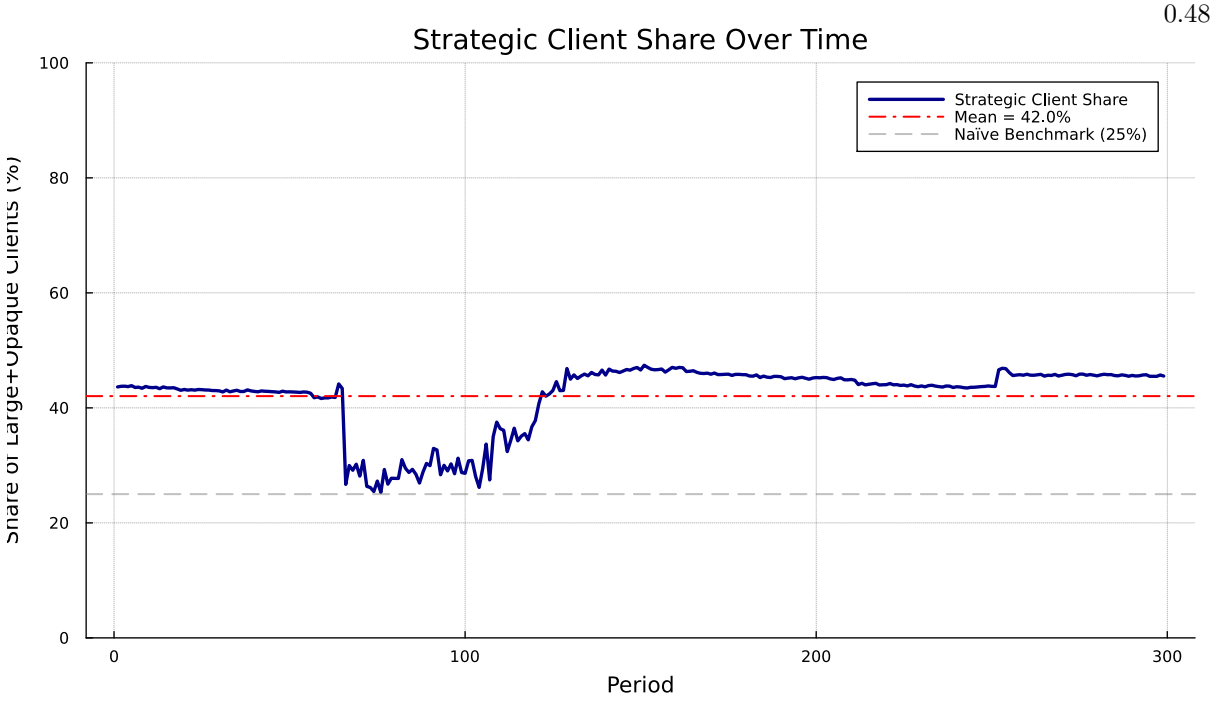


Figure 9: Client Mix Over Time

Figure 10: Dynamics of Strategic Leniency (RQ4). Panel A: The strategic premium (excess inflation for Large, Opaque issuers) persists dynamically. Panel B: The market share of Large, Opaque issuers remains elevated, reflecting endogenous selection.

Figure 11: The Endogenous Cycle of Trust and Leniency (RQ4). Panel A: Aggregate Market Trust (κ_t). Panel B: Average CRA Strategic Posture. Panel C: Average Individual CRA Reputation (R_t). Panel D: High-Grade Default Rate.